

Channel Models

Ack: Kameswari Chebrolu

Channel Models

- Wireless channels are extremely random and not easy to analyse
 - Models often statistical based on measurements
- Why are such models needed?
 - Estimate radio coverage
 - Evaluate performance of modulation techniques
 - Understand how parameters such as central frequency, bandwidth, mobile speed etc affect communication systems

Channel Models

- Three types of channel models and combinations employed in practice
 1. Additive White Gaussian Noise (AWGN)
 2. Large scale propagation model
 3. Small scale propagation model

Every model makes assumptions and not as accurate
as real experiments

Additive White Gaussian Noise (AWGN)

- Simplest of all channel models: considers noise alone
 - Additive: linear addition to intended signal
 - White: constant spectral density (like the colour white)
 - Gaussian: Normal/Gaussian distribution in time domain (average is zero)

Additive White Gaussian Noise (AWGN)

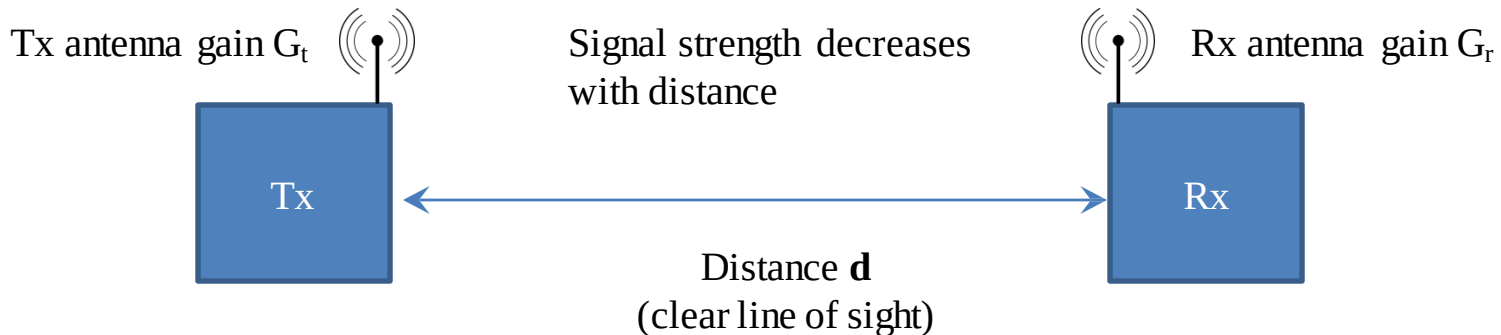
- Not a very good model for terrestrial links (due to multipath effects)
 - Used in conjunction with other models

Large Scale Propagation Models

- Predict average RSS (received signal strength) at a given distance from transmitter (TX)
- Distances range from few hundreds of meter to kms
- Useful in estimating radio coverage
- Cover some popular Path Loss models
 - Free Space, Two-Ray, Log-distance, Shadowing, Statistical models

Free Space Propagation

- Clear unobstructed line of sight between transmitter and receiver
 - Earth surface, buildings, vegetation etc ignored
 - E.g. satellite communication, microwave LOS (line-of-sight) communication



Free Space Propagation

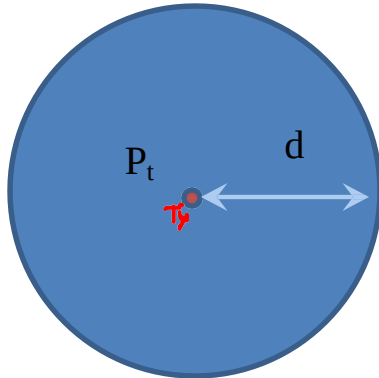
- Friis free space equation:

$$Loss = P_t/P_r = (4\pi d)^2 / (G_t G_r \lambda^2)$$

$$P_r = [(G_t P_t) / (4\pi d^2)] * [(G_r \lambda^2) / 4\pi]$$

Power flux density of of transmission .

Receiver's effective area



d : distance between Tx and Rx

G_t : transmitter antenna gain

G_r : receiver antenna gain

wavelength $\lambda = c/f$

where c : speed of light, f : carrier frequency

Decibels

- A dimension-less quantity that expresses the ratio between two power quantities in logarithmic scale
 - $10\log_{10} (P1/P2)$
- Example: 3dB attenuation means $P2 = P1/2$ i.e. power reduced by half
- dBm (decibel-milliwatts): Used to measure power in dB relative to “1 milliwatt”
 - 0dBm means 1 milliwatt of power; what is 20dBm?

$$P_r = [(G_t P_t)/(4\pi d^2)] * [(G_r \lambda^2)/4\pi]$$

$$\text{PL (in dB)} = 10 \log_{10}(P_t/P_r)$$

$$P_r = P_t + G_t + G_r - L_p \text{ (all terms in dB or dBm)}$$

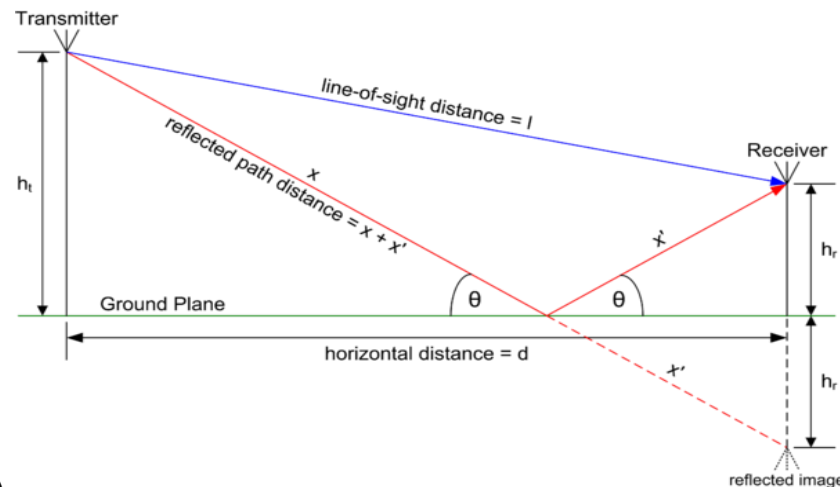
$$\text{Free space isotropic antenna loss} = L_p \text{ (in dB)}$$

$$L_p = 20 \log f + 20 \log d - 147.56$$

Ray Tracing Models

- Consider finite number of reflectors with known locations and dielectric properties
- Apply Maxwell's equations (for electromagnetic waves) to determine received power
- Very complex and not very accurate

Two Ray Ground Reflection Model



(Applicable for very large d)

$$P_r = (P_t G_t G_r h_t^2 h_r^2) / d^4$$

PL (in dB) =

$$40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

$$\text{PL (in dB)} = 10 \log_{10}(P_t/P_r)$$

Free Space

$$P_r = P_t + G_t + G_r - L_p \text{ (all terms in dB or dBm)}$$

$$\text{Free space isotropic antenna loss} = L_p \text{ (in dB)}$$

$$L_p = 20 \log f + 20 \log d - 147.56$$

(Applicable for very large d)

$$P_r = (P_t G_t G_r h_t^2 h_r^2)/d^4$$

Two Ray Model

$$\text{PL (in dB)} =$$

$$40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

Comparison

- Free Space vs Two-Ray Model
 - Frequency dependent vs independent
 - d^2 vs d^4 i.e 20dB/decade vs 40 dB/decade
 - Antenna height independent vs dependent

Minor change in model,
Major difference in end-result

Statistical Models

- Mobile systems operate in complex environments
 - Free-space/Ray tracing models not accurate
- Empirical Path Loss Models: Measure P_r/P_t over a given
 - Environment (urban, rural, inside buildings)
 - Distance
 - Frequency range
 - P_r/P_t measurement includes path loss, shadowing, multipath effects

- For a given environment (say dense urban), to determine $PL(d)$, conduct experiments
 - Over several wavelengths at distance d ,
 - Vary location of transmitter, receiver in the environment but maintain same distance d
 - Vary location (similar environment. E.g. South Mumbai, Connaught Place)
 - $PL(d)$: average of all measurements in the environment
- Examples: Okumura, Hata, Indoor

Simplified Path Loss Model

- Accurate path loss models require complex analysis and empirical measurements
 - E.g. Cellular networks employ them for placement of base-stations
- For other purposes (e.g. research) , a simple approximate model may suffice

$$P_r = P_t K (d_0/d)^\gamma$$

$$P_r \text{ (in dBm)} = P_t \text{ (in dBm)} + K \text{ (in dB)} - 10\gamma \log(d/d_0)$$

K, γ can be set based on an analytical or empirical model

d_0 is a reference distance

K captures average channel attenuation.

Analytical: free space path gain at d_0

Empirical: Field measurement at d_0

γ is path-loss exponent dependent on environment

Analytical: set to 2 or 4 (free space or 2-ray model)

Empirical: derived from field measurements

Environment	γ Range
Urban Macrocells	3.7 to 6.5
Urban Microcells	2.7 to 3.2
Office Building (same floor)	1.6 to 3.5
Office Building (multiple floors)	2 to 5.2
Store	1.8 to 2.2
Factory	1.6 to 3.3

Shadowing

- Same distance from transmitter, received power could be vastly different
 - Why? Different level of clutter in the path
- Common model: log-normal shadowing (tested empirically as well)

$$PL(d) \text{ (in dB)} = \bar{P}L(d) + X_\sigma$$

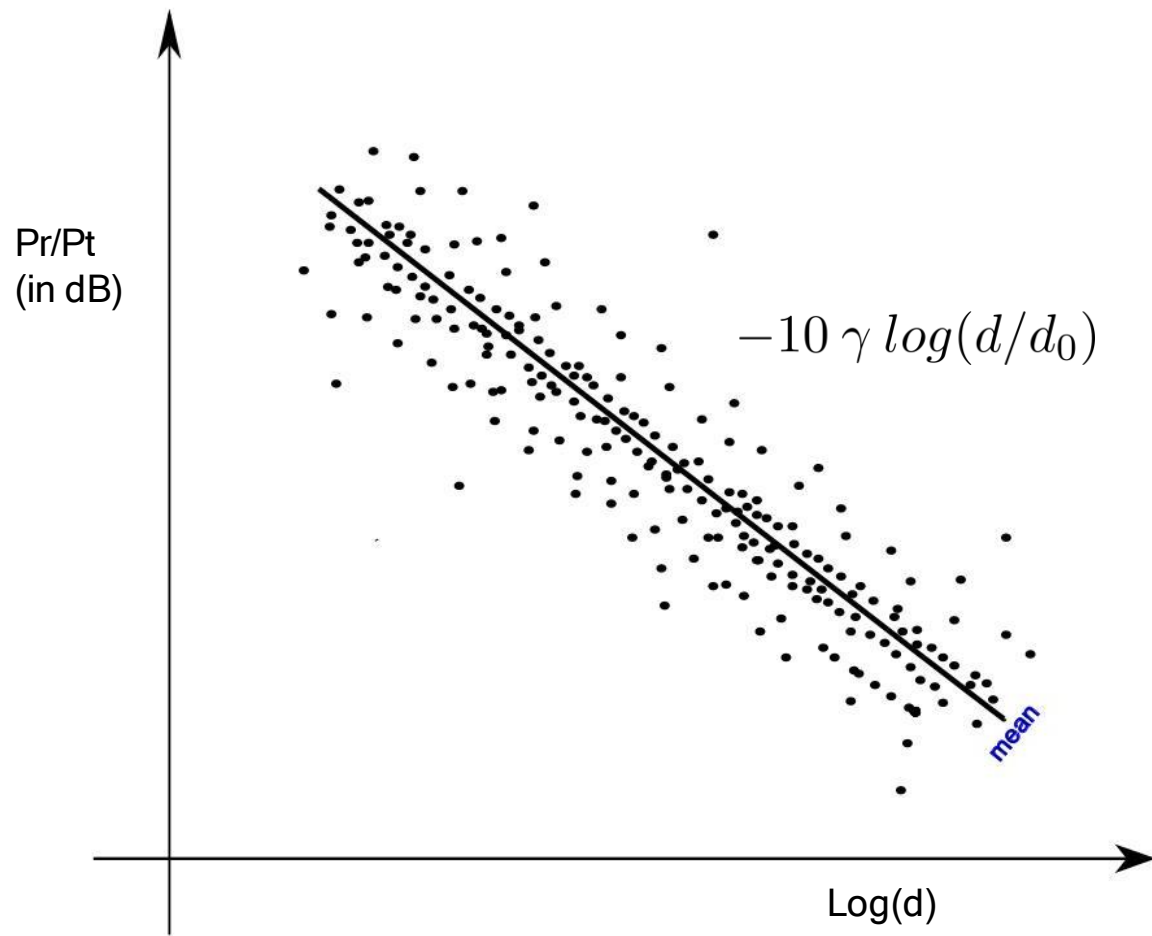
$$PL(d) \text{ (in dB)} = \bar{P}L(d_0) + 10\gamma \log(d/d_0) + X_\sigma$$

$$P_r(d) \text{ (in dBm)} = P_t \text{ (in dBm)} - PL(d) \text{ (in dB)}$$

X_σ : zero mean Gaussian random variable (in dB)

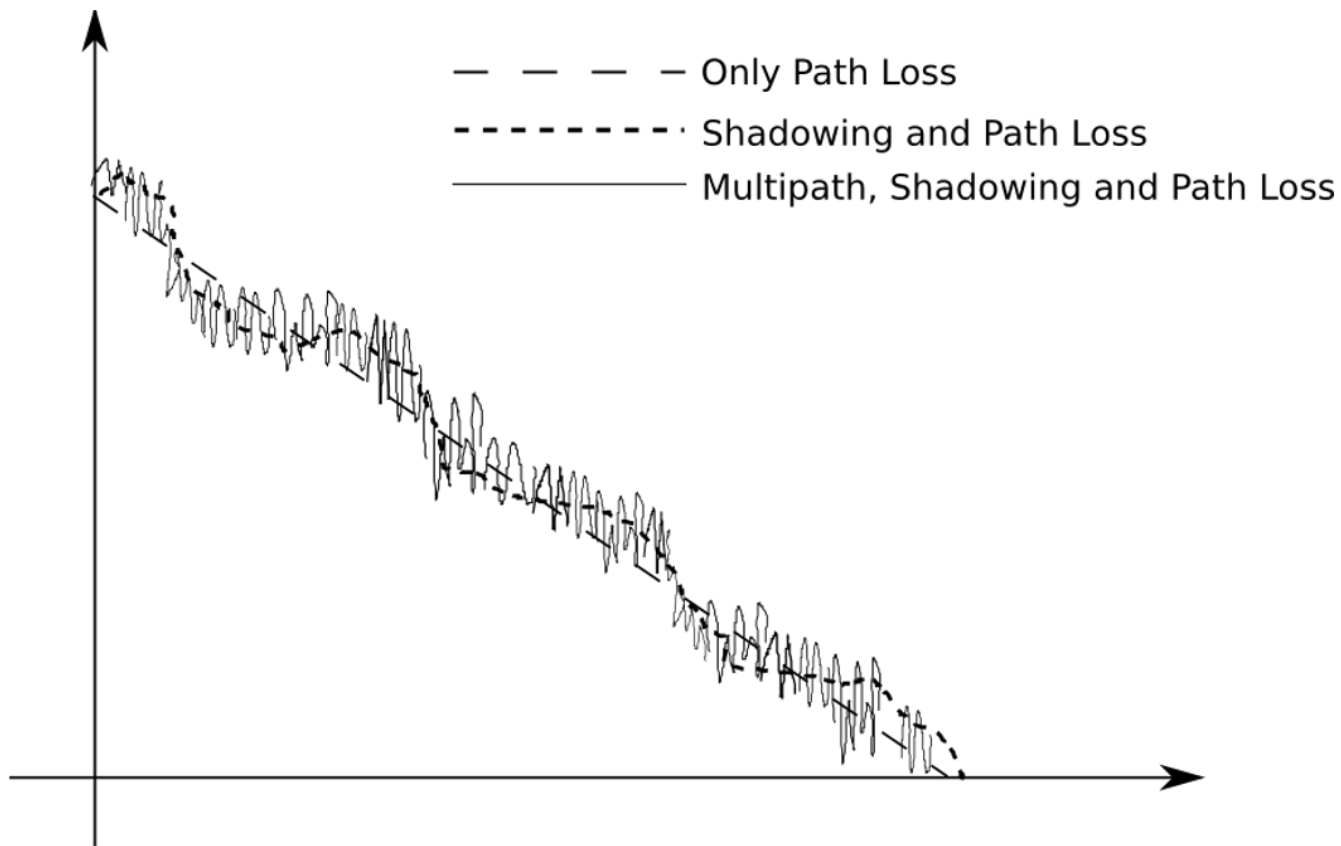
with standard deviation σ

(Typical values of σ 6 to 12 dB)



Small Scale Propagation Models

- Also called Fading models
- Characterize the rapid fluctuation of signal over
 - very short distances (order of wavelength ($\sim$ cms))
 - short time durations (us to ms)

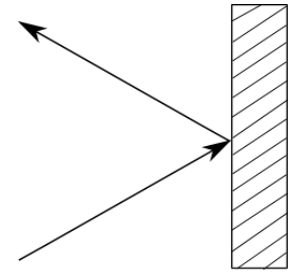
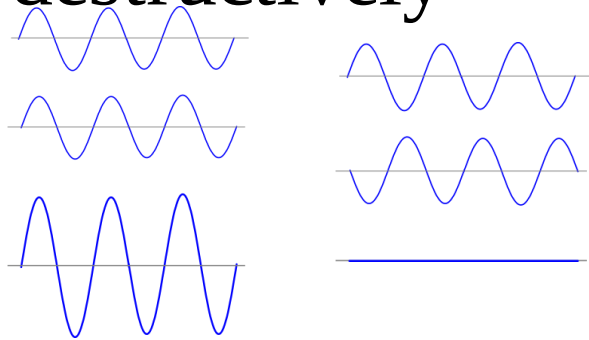


Outline

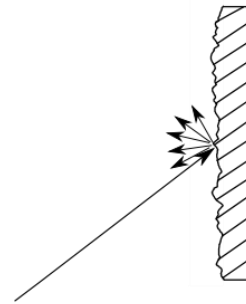
- **What causes fading?**
- How to model fading?
- Fading channel parameters
 - Parameters: Delay spread, coherence bandwidth and Doppler spread, coherence time
 - Types of fading: flat, frequency-selective and slow, fast fading

Fading/Multipath

- Apart from LOS, signal can reach receiver in other ways
 - Reflection, Diffraction and Scattering
- Multiple copies add constructively or destructively



Reflection



Scattering



Diffraction

End Result: Fading

- Rapid changes in the signal strength
 - In space as well as time
- Frequency modulation due to Doppler shift (mobility induced)
- Inter-symbol interference (due to multipath propagation delays)

Factors Influencing Fading

- Amount of clutter in environment (multipath propagation)
- Speed of the receiver/transmitter or even surrounding object (Doppler shift)
- Bandwidth of the signal

Characterization

- Fading is random
 - random amplitudes, phases and delays of the random multipath components
- Characterized statistically or via measurements
- Statistically: Assumed to be a complex Gaussian process
 - Rayleigh, Ricean, Clarke's fading models